ML: MULTIPLE CHOICE QUESTIONS - 1

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1. In supervised learning, the training data contains: Answer :C
   A) Only input data
   B) Only output data
   C) Input data with corresponding labels
   D) Random values

2. Which of the following is an example of supervised learning? Answer :B
   A) Clustering
   B) Classification
   C) Dimensionality reduction
   D) Association rule mining

3. Classification is used when the output variable is: Answer :B
   A) Continuous
   B) Discrete/Categorical
   C) Random
   D) Missing

4. Regression is used when the output variable is: Answer :C
   A) Binary
   B) Categorical
   C) Continuous
   D) Nominal

5. Which algorithm is commonly used for classification? Answer :B
   A) K-Means
   B) Decision Tree
   C) PCA
   D) Apriori

6. Which of the following is a binary classification problem?
   Answer :B
   A) Predicting house price
   B) Detecting spam emails
   C) Forecasting sales
   D) Predicting temperature

7. In multi-class classification, the target variable has: Answer :B
   A) Two classes only
   B) More than two classes
   C) No classes
   D) Infinite classes

8. Which metric is commonly used for classification evaluation?
   Answer :A
   A) Accuracy
   B) RMSE
   C) MAE
   D) MSE

9. Accuracy is defined as: Answer :A
   A) Correct predictions / Total predictions
   B) Incorrect predictions / Total predictions
   C) TP / FP
   D) FN / TP

10. Which of the following is NOT a classification algorithm?
    Answer :D
    A) Logistic Regression
    B) Naïve Bayes
    C) KNN
    D) Linear Regression

11. KNN stands for: Answer :A A) K-Nearest Neighbors
    B) Kernel Neural Network
    C) Known Node Network
    D) Key Number Node

12. In KNN, classification is based on: Answer :B
    A) Random guessing
    B) Nearest data points
    C) Largest value
    D) Mean output

13. Decision Tree splits data based on: Answer :A
    A) Rules/conditions
    B) Random values
    C) Labels only
    D) Noise

14. Logistic Regression is mainly used for: Answer :A
    A) Classification
    B) Clustering
    C) Regression only
    D) Sorting

15. Which algorithm is probabilistic? Answer :B
    A) SVM
    B) Naïve Bayes
    C) KNN
    D) Decision Tree

16. Which of the following is an application of classification?
    Answer :A
    A) Face recognition
    B) Data compression
    C) Noise reduction
    D) Sorting files

17. Overfitting occurs when a model: Answer :A
    A) Performs well on training but poorly on test data
    B) Performs poorly on all data
    C) Has fewer features
    D) Uses small datasets only

18. Underfitting occurs when a model: Answer :B

A) Learns noise
B) Cannot capture patterns properly
C) Memorizes data
D) Has too many layers

19. Which dataset split is used to evaluate final performance?
Answer :C

A) Training set
B) Validation set
C) Test set
D) Input set

20. Confusion matrix is used for: Answer :B

A) Regression
B) Classification evaluation
C) Clustering
D) Data storage

21. Precision measures: Answer :A

A) Correct positive predictions among predicted positives
B) Correct negatives only
C) Total predictions
D) Error rate

22. Recall measures: Answer :A

A) Correct positives among actual positives
B) Correct negatives
C) Total outputs
D) Accuracy only

23. Which algorithm uses hyperplane separation? Answer :A

A) SVM
B) KNN
C) Naïve Bayes
D) PCA

24. A model predicting Yes/No outcomes is an example of:

Answer :B

A) Regression
B) Binary Classification
C) Clustering
D) Reinforcement Learning

25. Which of the following is supervised learning application?

Answer :A

A) Email spam filtering
B) Market basket analysis
C) Customer segmentation
D) Topic modeling